Railway level crossing in mCRL2

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Introduction

This report was written for the first assignment of the *Systems Verification* course at *Faculdade de Ciências da Universidade do Porto* (FCUP). It presents a model of a railway level crossing, describing the interaction between a controller, a gate, and one or more trains that wish to cross. We then formalise a set of properties that verify its correctness and check them in mCRL2.

Task 1

Components

Composition

Inspection

Properties

Broken Variant

Task 2

Components

Composition

Inspection

Properties

Broken Variant

Conclusion